CPSY 200 - Software Analysis and Design

Description
This course introduces the theory and practice of software development, focusing on the requirements gathering, analysis and design phases of the software development life cycle. You'll also learn how computer-aided software engineering (CASE) programs can aid in the analysis and design processes.

3 Credits

Time Guidelines
The standard instructional time for this course is 60 hours.

Prerequisites

- CPRG 216

Course Assessment

Assignments	40%
Project(s)	60%
Total:	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Other Course Information
Learner Engagement:

In order to be successful, the learner is expected to be engaged in learning activities for a total of 9 to 12 learning hours per course per week, which includes both in-class and out-of-class time.

Course Learning Outcomes
1. Describe the different phases the software development lifecycle (SDLC).

Objectives:

1.1 Describe information systems using systems theory concepts

1.2 Describe the roles in the systems development team.

1.3 Explain the various software development lifecycle frameworks.

1.4 Explain the different types of requirement gathering techniques including the various types of requirements

1.5 Demonstrate how to conduct a client interview to develop a list a business requirement

1.6 Summarize the elements of the Unified Modelling Language (UML).

2. Construct a use case diagram.

Objectives:

2.1 Describe the different components of a use case diagram.

2.2 Describe the features of a case tool.

2.3 Use a case tool to document components of a system.

2.4 Create a sequence diagram and descriptions for a use-case

2.5 Explain the difference components of a sequence diagram

3. Produce a data dictionary for a system.

Objectives:

3.1 Classify different types of data.

3.2 Describe the rules for valid values.

3.3 Apply cardinality and optionality to data elements.

4. Produce a class diagram for a given system.

Objectives:

4.1 Describe the different components of a class diagram.

4.2 Identify components from system requirements.

4.3 Categorize types of UML relationships.

4.4 Compare inheritance, encapsulation and polymorphism.

5. Develop operation specifications from requirements and class diagrams.

Objectives:

5.1 Describe Structured English.

5.2 Write operation specifications for a process using Structured English.

5.3 Compare algorithmic and non-algorithmic specifications.

5.4 Describe various techniques of operation specifications.

5.5 Describe the different components of an activity diagram.

6. Produce a state machine diagram.

Objectives:

6.1 Describe time dependent behaviors

6.2 Outline the impacts of time dependent behaviors on design.

6.3 Relate time dependent behaviors to state diagrams.

6.4 Describe the different components of a state diagram.

6.5 Illustrate time dependent behaviors using a state diagram.

7. Incorporate design patterns into system diagrams.

Objectives:

7.1 Describe types of design patterns.

7.2 Apply usability standards to interface design.

7.3 Illustrate a prototype.

7.4 Integrate interface and database objects into class diagrams.

7.5 Integrate interface and database objects into sequence diagrams.

SAIT Policies and Procedures:
Students should familiarize themselves with SAIT's Policies and Procedures, including but not limited to those in the Academic Student, Conduct, and Health, Safety, and Environment domains.

Copyright 2015 - 2025, Southern Alberta Institute of Technology (SAIT). All Rights Reserved.

This document and materials herein are protected by applicable intellectual property laws. Unauthorized reproduction and distribution of this publication in whole or part is prohibited.